AI FOR CREATIVITY · RESEARCH SKILLS WORKSHOP

From curiosity to a credible R&D document

Research gaps · problem statements · datasets · UCA Harvard referencing

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WHY THIS MATTERS

Your R&D document is one third of your portfolio

R&D document

Evidences your learning journey: research, data, experiments, ethics, reflection

AI model

Code commented to show the logic of your work

Essay or video essay

Places your practice in a critical framework

LO1 Detailed knowledge of contexts, concepts, technologies, skills, and processes

Research & analysis · Engaging with practice

LO3 Consistent, refined, and ethical application of advanced programming and machine learning

Experimentation & innovation

WORKSHOP PLAN · 2 HOURS

How the two hours run



0:00
Why R&D
What an R&D document is and how it is assessed

0:10
Find & read
Search strategy, reading papers, judging sources · Ex 1–2

0:30
Gaps & questions
Gap types, evidence, problem statement, RQs · Ex 4–5

1:00
Break

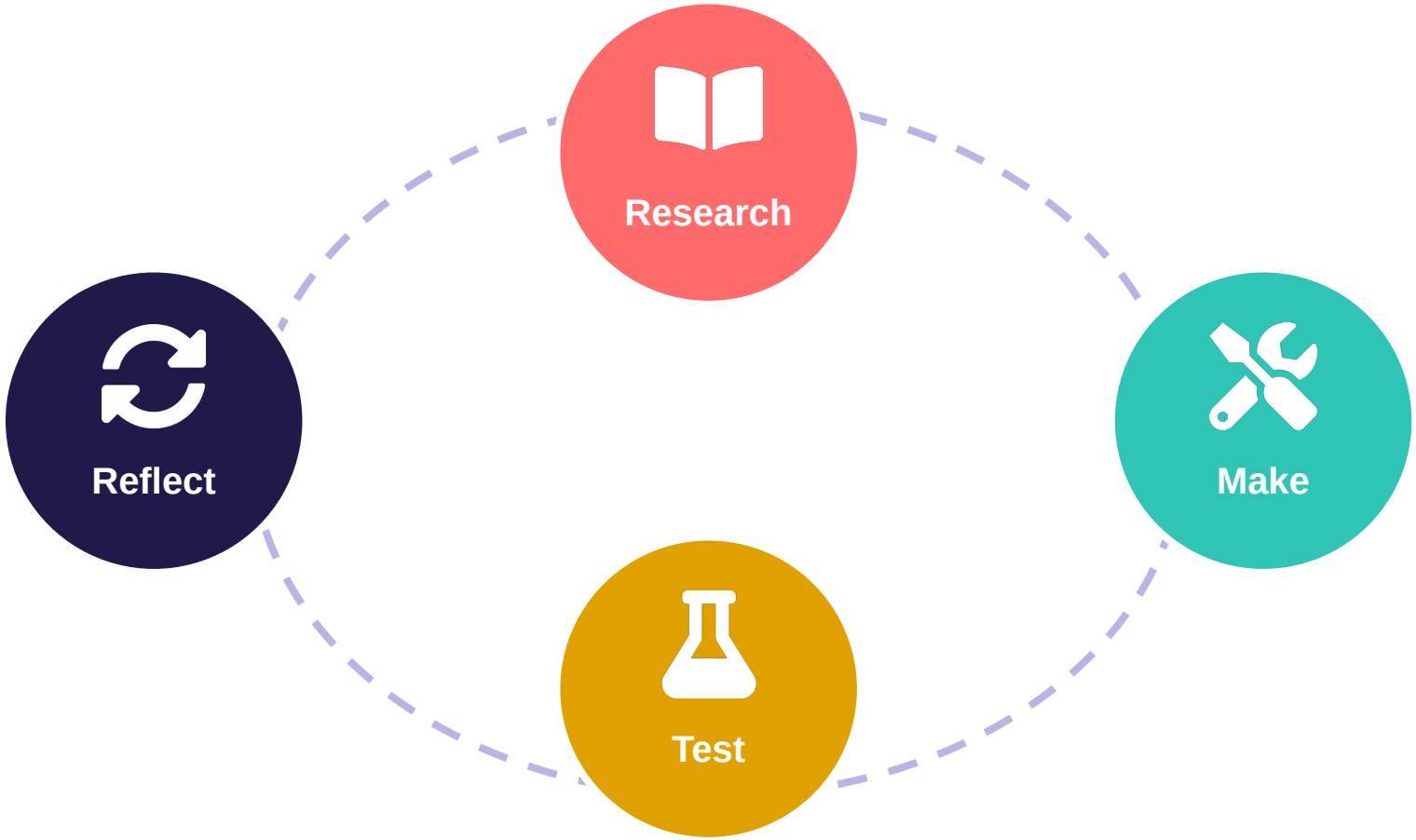
1:10
Datasets
Collecting, evaluating and documenting data · Ex 6–7

1:35
UCA Harvard
Citing and referencing, including datasets and AI · Ex 8–9

1:55
Wrap-up
Exemplar walk-through task and checklist

WHAT IS R&D?

In creative AI, R&D is a loop, not a line



Your R&D document records every lap

- What you read and why
- What you tried — including failures
- How you measured it
- What you changed and why
- How your thinking changed

THE RESEARCH JOURNEY

Eight steps from curiosity to contribution

1 Curiosity

A question from your practice

2 Context

Why does it matter, and to whom?

3 Literature

What is already known?

4 Gap

What is missing — with evidence

5 Problem

Who is affected, and how?

6 Questions

What exactly will you find out?

7 Data & method

How will you find out, ethically?

8 Reflection

What did you learn? What next?

Today's focus: the dark boxes — the gap, the problem and your data — are where most student documents are weakest.

01

Finding and reading sources

Search like a researcher, read like a critic.

WHERE TO LOOK

Six places good sources live

UCA library search

Books, e-books and journals you can access in full. Start here.

Google Scholar

Wide academic search. Use 'Cited by' to find newer work on the same topic.

arXiv · CVF · ACM · IEEE

AI and computer-vision papers, often free (CVF Open Access, arXiv).

Museums & heritage bodies

UNESCO, museum collections and archives for cultural context.

Industry reports

Trend and market reports (e.g. WGSN) — useful context, but check their purpose.

Dataset hubs

Hugging Face, Kaggle, UCI, museum open-access collections — always check the licence.

SEARCH STRATEGY · WORKBOOK EX 1

Turn your idea into a search string

Concept 1: the craft	Concept 2: the technique	Concept 3: the problem
`"Al Sadu"  OR`	`"deep learning"  OR`	`authentic*  OR`
`"Sadu weaving"  OR`	`"computer vision"  OR`	`counterfeit  OR`
`"Bedouin textile"`	`"image classification"`	`imitation`

```
("Al Sadu" OR "Sadu weaving" OR "Bedouin textile") AND ("deep learning" OR "computer vision" OR "image classification") AND (authentic* OR counterfeit OR imitation)
```

Rules: OR between synonyms · AND between concepts · "quotes" for exact phrases · * for word endings · log every search with its date.

READING PAPERS

Read a paper in three passes, not one

1 Pass 1 · 5–10 min

Title, abstract, headings, figures, conclusion. Decide: is it relevant?

2 Pass 2 · up to 1 hour

Read the whole paper, skip the maths. Note the method, data, results and limitations.

3 Pass 3 · only if essential

Work through details — you probably need this for 2–3 key papers only.

For every source, note 4 things: What did they do? · With what data? · What did they find? · What did they NOT cover (for my project)?

JUDGING SOURCES · WORKBOOK EX 2

The CRAAP test: is this source good enough?

C

Currency

Recent enough for a fast-moving AI topic?

R

Relevance

About my question, or only the general topic?

A

Authority

Peer-reviewed? Who are the authors?

A

Accuracy

Evidence, data, references — can I verify it?

P

Purpose

To inform, to sell or to persuade?

Try it: a company blog claiming 'AI saves designers 80% of their time' — how does it score on Purpose and Accuracy?

WRITING THE REVIEW

Summarise less, synthesise more

✗ Summary (weak)

“Cimpoi et al. (2014) made a texture dataset. Yosinski et al. (2014) studied transfer learning. Howard and Gugger (2020) wrote a book about fastai.”

A list. No connection. No 'so what'.

✔ Synthesis (strong)

“Textures are learnable (Cimpoi et al., 2014), and pretrained features transfer to new tasks (Yosinski et al., 2014), which makes a small craft dataset realistic. However, none of these studies includes Gulf textiles.”

Connects sources → implication → gap.

Tool: the literature matrix (Workbook Ex 3) — one row per source; its last column, 'What it does NOT cover', is where your gap appears.

02

Research gaps and questions

What is missing, why it matters, and what exactly you will find out.

RESEARCH GAPS

A research gap = something missing that matters



Missing

Something is not yet known, tested, built or documented.



Evidenced

You can show it is missing: your search log, your matrix.



Matters

Filling it helps someone: a community, a practice, an industry.

Never write: “Nobody has ever done this.” **Write instead:** “In my search of X, Y and Z on 12/09/2026, I found no…”

SIX TYPES OF GAP

Name your gap: which type is it?

Context

Studied elsewhere, not here: Emirati textiles, Khaleeji music, Arabic typography.

Data

No suitable, licensed, consent-based dataset exists for this subject.

Method

Measured the wrong way — e.g. clicks instead of creative usefulness.

Application

The technique exists but has not been applied in this practice.

Evidence

Claims exist (often from companies) but independent evidence does not.

Ethical

Consent, credit, payment or cultural impact is under-examined.

EVIDENCE YOUR GAP

Show your search, don't just claim the gap

Search string	Screened	Relevant	Finding
`"textile classification" AND "deep learning"`	50	14	Mostly industrial defects, garment types
`"Al Sadu" AND ("machine learning" OR "computer vision")`	9	0	Cultural and historical studies only
`"handwoven" AND ("counterfeit" OR "imitation") AND "image"`	23	2	South Asian handlooms; no Gulf textiles

Result → three named gaps: context (Gulf textiles missing) · data (no Sadu dataset) · application (hand-made vs printed not studied for Sadu).

From the annotated exemplar, Section 3. Search counts are illustrative, invented for teaching.

ACTIVITY · SPOT THE GAP · WORKBOOK EX 4

Which type of gap? Pairs, 5 minutes

1. Most music-generation studies train on Western classical and pop; very few include Khaleeji or Arabic maqam music.

2. AI fashion recommenders are evaluated on clicks, never on whether designers found the suggestions creatively useful.

3. Papers on AI voice cloning rarely discuss how the original performers consented or were paid.

Choose from

- Context
- Data
- Method
- Application
- Evidence
- Ethical

Then: write one gap statement for your own project using the 'In my search of… I found no…' pattern.

FROM GAP TO PROBLEM · WORKBOOK EX 5

The problem statement formula

WHO is affected	+ WHAT is wrong	+ WHY existing work doesn't solve it
Weavers and buyers of Al Sadu	*Printed imitations are sold alongside hand-woven pieces; buyers can't tell them apart*	*No texture research or dataset includes Emirati textiles*

Aim (one sentence, starting 'To...')

To explore whether transfer learning on a small, ethically collected image dataset can distinguish hand-woven Al Sadu from machine-printed imitations, and to document the dataset and process so others can build on it.

RESEARCH QUESTIONS

Sharpen your research questions

✘ **Can AI help culture?** *Too broad*	✔ How accurately can a fine-tuned CNN classify hand-woven Sadu, printed Sadu-style and other woven textures on unseen objects?
✘ **Can AI recognise Sadu?** *Yes/no only*	✔ Which visual features does the model rely on, and where does it fail?
✘ **Will my app stop counterfeiting in the UAE?** *Not answerable in 14 weeks*	✔ What ethical and practical conditions must be met to collect and share a craft dataset responsibly?

Test each RQ: specific? answerable with my data and method? achievable by Week 14? more than yes/no?

OBJECTIVES

Objectives: SMART steps towards your aim

Collect and document ≥350 images per class with written consent	by Week 8
Train and evaluate a baseline model	by Week 10
Improve the model and analyse its errors	by Week 12
Produce a datasheet, model card and Gradio prototype for user testing	by Week 13

SMART: Specific · Measurable · Achievable · Relevant · Time-bound. Your objectives become your project timeline.

03

Datasets

Where your data comes from, whether you may use it, and how to describe it.

SOURCES OF DATA

Four ways to get data — one to avoid



Collect your own

Photos, recordings, sketches. Most original; you control consent and quality. Takes time.



Public datasets

Hugging Face, Kaggle, UCI, museum open access. Fast — but check licence, bias and how it was made.



Synthetic data

Generated or augmented data. Useful for balance; must be labelled clearly as synthetic.



Scraped without permission

Instagram, Pinterest, Google Images. Legal, ethical and cultural risks. Avoid.

EVALUATE A DATASET · WORKBOOK EX 6

Seven questions to ask of any dataset

1. Who created it, when and why?
2. What is in it — classes, counts, format?
3. How was it collected? Did people consent?
4. What licence applies? Can I train on it?
5. How was it labelled, and by whom?
6. Who or what is missing or under-represented?
7. Could items leak between train and test?

Based on 'Datasheets for datasets' (Gebru et al., 2021)

DESCRIBE YOUR DATASET · WORKBOOK EX 7

Your datasheet: eleven fields

Name & version	Purpose	Classes & counts	**Sources & permissions**
Collection method	Labelling & checks	Pre-processing	**Splits (no leakage)**
Licence & sharing	**Limitations & bias**	Storage & privacy	**Dark = where most marks are lost**

Goes into: Section 5 of your R&D document. See the completed example in the annotated exemplar.

EXEMPLAR DATASHEET

Specific beats vague

1,230	58	3	58/60
images in total	hand-woven pieces	classes	labels confirmed by a weaver

✘ Vague

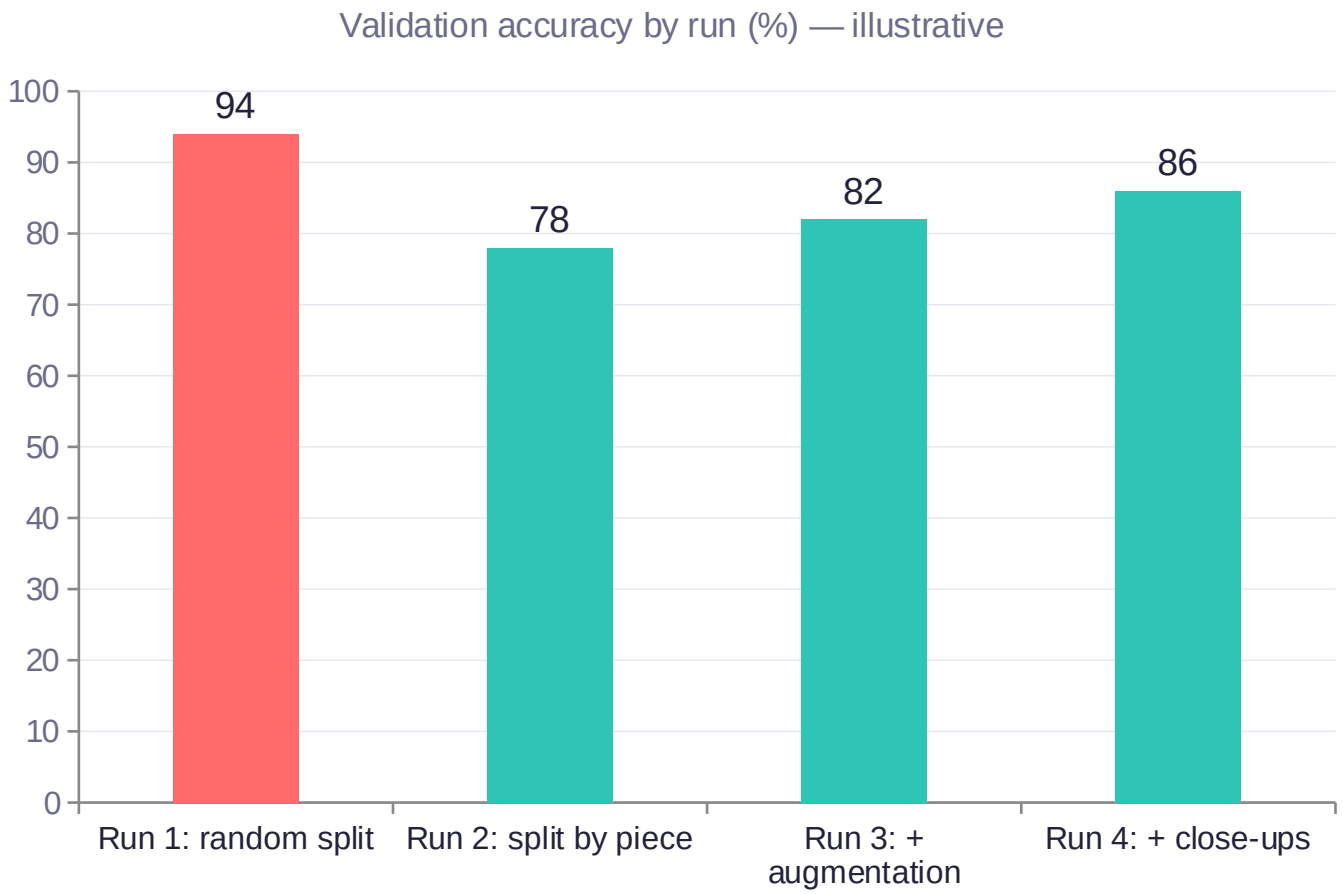
“I used a few hundred images from the internet of Sadu and other fabrics.”

✔ Specific

“400 images of 58 hand-woven pieces, photographed with written permission; split by piece, not by photo; licensed CC BY-NC 4.0.”

A LESSON FROM THE EXEMPLAR

When 94% was the wrong answer



Data leakage

Run 1 split photos randomly, so photos of the same textile landed in both training and validation. The model memorised pieces.

Splitting by piece gave an honest 78% — then real improvements followed.

DATASET RED FLAGS

Stop and rethink if you see…

No licence

If you can't find the licence, you can't assume you may use it.

Scraped social media

Creators never agreed; images may be copyrighted or personal.

Faces without consent

Personal data. Needs explicit consent — or leave faces out.

Heavy class imbalance

900 of one class, 40 of another: accuracy becomes misleading.

Near-duplicates

Burst photos or crops of the same object inflate results.

Sacred or restricted material

Some cultural content should not be used without community agreement.

04

UCA Harvard referencing

Credit other people's work — every time, the same way.

IN-TEXT CITATIONS

In the text: author, year, page

Paraphrase	`(Boden, 2004)`
Direct quotation	`(Butler, 2006:8)`
Two authors	`(Howard and Gugger, 2020)`
Three or more	`(Gebru *et al.*, 2021)`
Author in your sentence	`Boden (2004) argues that…`
Secondary citation	`(Lovelace, 1843 cited in du Sautoy, 2019:12)`

UCA style: colon before the page number (2006:8); et al. in italics for three or more authors; full stop after the bracket at the end of a sentence.

REFERENCE LIST ANATOMY

In the reference list: every part has a place

Author | (Year) | Title | Source details | Online: At: URL (Accessed)

Book

Boden, M.A. (2004) *The creative mind: myths and mechanisms.* (2nd ed.) London: Routledge.

Journal article (online)

Gebru, T., Morgenstern, J., ... and Crawford, K. (2021) 'Datasheets for datasets' In: *Communications of the ACM* 64 (12) pp.86-92. At: https://doi.org/10.1145/3458723 (Accessed 23/09/2026).

In your reference list, list every author; '...' is used here only to fit the slide.

WEBSITES · DATASETS · AI

Referencing websites, datasets and AI conversations

Website

UNESCO (2025) *Al Sadu, traditional weaving skills in the United Arab Emirates*. At: https://ich.unesco.org/en/RL/al-sadu-traditional-weaving-skills-in-the-united-arab-emirates-02223 (Accessed 23/09/2026).

Dataset

Cimpoi, M., Maji, S., Kokkinos, I., Mohamed, S. and Vedaldi, A. (2014) *Describable Textures Dataset (DTD)*. [Dataset] At: https://www.robots.ox.ac.uk/~vgg/data/dtd/ (Accessed 23/09/2026).

AI conversation

Open AI (2026) *AI conversation with Surname, Initial*. DD/MM/YYYY. At: [shareable URL]

Note: the UCA guide has no dataset format, so this adapts its website format with a [Dataset] label — confirm with your tutor. AI conversations follow the UCA personal-communication format.

ACTIVITY · FIX IT · WORKBOOK EX 9

What's wrong with these references?

1. `The creative mind, Margaret Boden, Routledge, 2004`

2. `Howard, J. & Gugger, S., Deep Learning for Coders (O'Reilly)`

3. `https://www.robots.ox.ac.uk/~vgg/data/dtd/`

Look for: author order · year in brackets · italics · place: publisher · 'and' not '&' · a bare URL is not a reference.

ACADEMIC INTEGRITY

Using AI tools honestly in your R&D

Allowed with declaration

- Explaining an error message
- Suggesting search keywords
- Checking grammar

Not acceptable

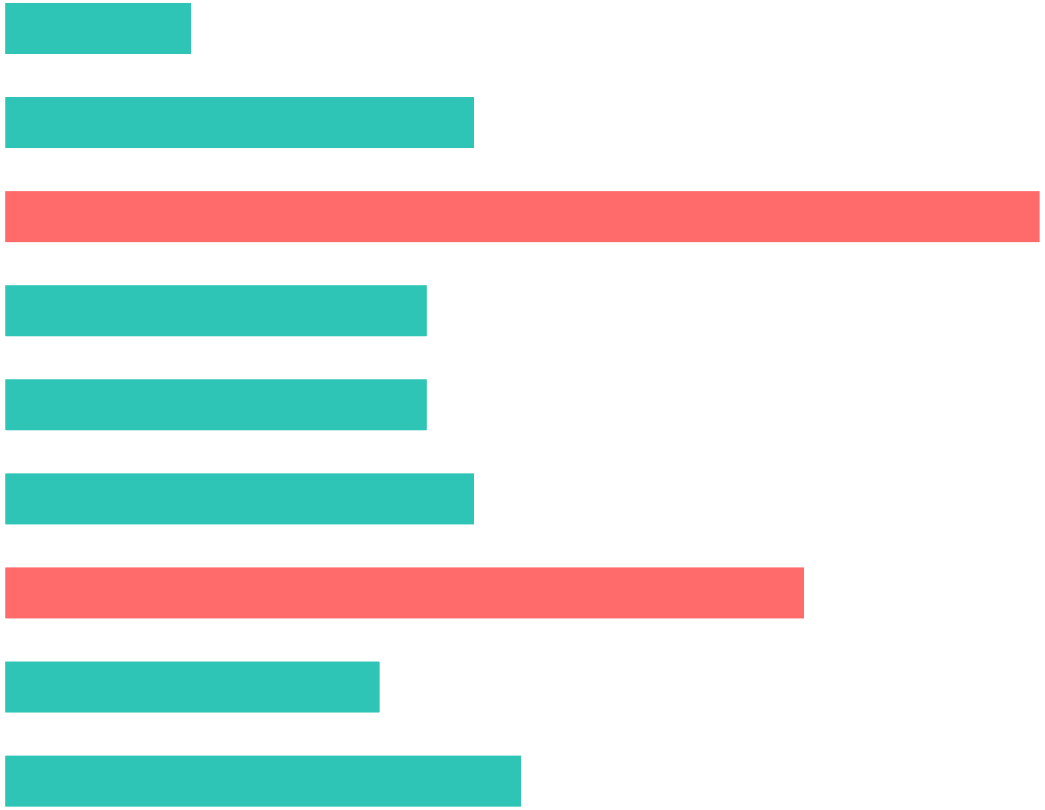
- Writing your sections for you
- Inventing sources or data
- Code you cannot explain

Beware: AI chatbots can invent references that look real. Check every source exists before you cite it — and always read it yourself.

STRUCTURE

A suggested structure for your R&D document

Abstract	1 paragraph
1 Introduction & context	~300 words
2 Literature & practice review	~700 words
3 Research gap	~250 words
4 Problem, aim, RQs, objectives	~250 words
5 Datasheet	table
6 Method & development log	~500 words + table
7 Ethics	~250 words
8 Reflection & next steps	~300 words
References + appendices	not counted



USING THE EXEMPLAR

Read the annotated exemplar like an assessor

Green boxes

Why a section works — copy the structure, not the content

Red boxes

Common mistakes — check your own draft against each one

Task (pairs, independent study)

- Find the three gap types in Section 3
- Find two limitations the student admits in the datasheet — why is that a strength?
- Check three in-text citations against the reference list
- Write one sentence: what will you do differently in your own R&D document?

BEFORE YOU SUBMIT

R&D document checklist

- [ ] Every factual claim is cited
- [ ] Gap is named by type and evidenced by a search log
- [ ] Problem → aim → RQs → objectives connect
- [ ] Datasheet has numbers, sources, licence, consent, splits, limitations
- [ ] Development log shows failures and changes
- [ ] Ethics section is specific to my project
- [ ] Reflection says what I learned and what next
- [ ] Reference list matches in-text citations (UCA Harvard)
- [ ] AI use is declared

Good research is honest research.

Show your search. Name your gap. Describe your data. Credit your sources.

Before next week:
Finish Workbook Exercises 3 and 7 · read the annotated exemplar · add your search log and first datasheet draft to your R&D journal